The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Germ-Generating Substrate Proto-Germ Dynamics
from Proto-Germ-Generating Substrate Pre-Proto-Germ Dynamics

Abstract

The active primitive-reservoir observer-reduction frontier is now the theorem deriving the current germ package from germ-generating substrate proto-germ dynamics. Under the recorded routing safeguard, the honest next benchmark-facing move must therefore do one of two things: either derive or justify the proto-germ package from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside the observer-relevant proto-germ package. The present note takes the default first route.

For each sector it introduces *proto-germ-generating substrate pre-proto-germ dynamics*: a compact convex pre-proto-germ body over the recorded proto-germ state space, unique pre-proto-germ minimizers on the fixed proto-germ, germ, seed, foundation, ground, origin, precursor, lift, and substrate fibers, a smooth pre-proto-germ restriction section, an exact pre-proto-germ shell projecting diffeomorphically to the exact proto-germ shell, a closed embedded pre-proto-germ zero core whose projected split reproduces the proto-germ split, a hidden charge, an observer-compatible exact pre-proto-germ zero-response realization, and a fixed-data solvability / coercive package projecting to the recorded proto-germ kernel data. The current germ-generating substrate proto-germ dynamics are then recovered canonically as the projected exact-shell transport image of that deeper pre-proto-germ package.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. The positive result is therefore narrow but benchmark facing: the current proto-germ package is no longer left as an unexplained primitive stopping point on the active line. The remaining honest burden moves one layer deeper again, to the pre-proto-germ dynamics themselves or to a sharper collapse inside their observer-relevant core.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has now reached a sharper burden than another same-output proto-germ restatement. The current proto-germ theorem already showed that the present germ package is the exact-shell transport image of a more primitive germ-generating substrate proto-germ dynamics package. Once that result is read together with the benchmark-routing safeguard against recursive depth drift, the next honest move becomes narrower: either derive or justify the proto-germ package from still more primitive substrate structure, or prove a genuinely smaller theorem-level collapse strictly inside the observer-relevant proto-germ package. Reopening the older root-nucleus handoff, re-running another same-output germ / seed / root / foundation relay, or returning to stationary-reduction, stationary-Hessian, spectral, or selector layers would no longer address the live burden.

The present note takes the first route. Its positive claim is not that final substrate microphysics has been identified, and it is not that the benchmark exact-relativistic package has been changed. Its claim is narrower and benchmark facing. The current proto-germ package is shown to arise canonically from a more primitive pre-proto-germ-level substrate dynamics whose projected exact-shell transport already contains the observer-relevant structure of the proto-germ package. In that sense the proto-germ layer is no longer the primitive place where the active line has to stop.

Two features matter here. First, the theorem targets the current proto-germ package itself rather than revisiting the already-spent germ handoff, so it does not spend the next move on an older same-output presentation. Second, the deeper pre-proto-germ package is not merely another renaming of the proto-germ package. It adds an explicit pre-proto-germ state space, pre-proto-germ body, pre-proto-to-proto projection, and deeper exact-shell transport data below the proto-germ frontier. The proto-germ shell, proto-germ zero core, split, and lower minimizer hierarchy are therefore projected images of a deeper substrate package rather than unexplained primitive data.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current germ-generating substrate proto-germ dynamics can itself be generated from a more primitive pre-proto-germ-level substrate dynamics.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded lower-fiber ladder)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes the lower-fiber ladder*

$$\mathcal{F}^{\mathrm{fdn}}_{\sigma}, \qquad \mathcal{G}^{\mathrm{grd}}_{\sigma}, \qquad \mathcal{O}^{\mathrm{orig}}_{\sigma}, \qquad \mathcal{P}^{\mathrm{prec}}_{\sigma}, \qquad \mathcal{L}^{\mathrm{lift}}_{\sigma}, \qquad \mathcal{M}^{\mathrm{sub}}_{\sigma},$$

together with the common substrate target $E^{\mathrm{sub}}_{\sigma}$, *the positive-definite pairing*

$$\mathfrak{s}^{\mathrm{sub}}_{\sigma} : E^{\mathrm{sub}}_{\sigma} \times E^{\mathrm{sub}}_{\sigma} \to \mathbb{R},$$

and the hidden equation target $Q^{\mathrm{eqn}}_{\sigma}$.

Definition 3 (Recorded downstream seed package)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already carries a recorded seed package* $\mathfrak{S}^{\mathrm{seed}}$, *consisting of:*

1. *a seed state space* $\mathcal{S}^{\mathrm{seed}}_{\sigma}$, *compact convex seed body* $\mathcal{B}^{\mathrm{seed}}_{\sigma}$, *affine seed projection* $\Pi^{\mathrm{seed}}_{\sigma} : \mathcal{S}^{\mathrm{seed}}_{\sigma} \to \mathcal{F}^{\mathrm{fdn}}_{\sigma}$, *and the fixed seed minimizers*

$$\mathfrak{f}^{\mathrm{seed,fdn}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{seed,grd}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{seed,orig}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{seed,prec}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{seed,lift}}_{\sigma}, \quad \mathfrak{m}^{\mathrm{seed}}_{\sigma};$$

2. *a seed restriction section* $\Gamma^{\mathrm{seed}}_{\sigma}$, *seed functional* $\mathscr{W}^{\mathrm{seed}}_{\sigma}$, *and exact seed shell* $\mathcal{Z}^{\mathrm{seed}}_{\sigma}$;

3. *a closed embedded seed zero core* $\mathcal{Z}^{0,\mathrm{seed}}_{\sigma}$, *hidden fiber space* $\mathcal{H}^{\mathrm{hid}}_{\sigma}$ *with compact hidden body* $D^{\mathrm{hid}}_{\sigma} \subset \mathcal{H}^{\mathrm{hid}}_{\sigma}$, *and hidden split diffeomorphism* $\Phi^{\mathrm{seed}}_{\sigma}$;

4. *the hidden charge* $\mathcal{Q}^{\mathrm{seed}}_{\sigma}$, *exact seed zero-response realization* $\zeta^{\mathrm{seed}}_{\sigma}$, *and fixed-data solvability / coercive package recorded by* $\widetilde{K}^{0,\mathrm{seed}}_{\sigma}$ *and* $P^{\mathrm{seed,co}}_{\sigma}$.

Remark 1. *Definitions 1–3 are not reopened here. The primitive continuation data, the one operative metric, the ordinary matter route, the recorded lower-fiber ladder, and the downstream seed package remain fixed inputs. The present note addresses only whether the current proto-germ package can itself be generated from more primitive substrate dynamics.*

3. Recorded Germ and Proto-Germ Targets

Definition 4 (Recorded seed-generating substrate germ dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The recorded germ package* $\mathfrak{G}^{\mathrm{germ}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean germ state space* $\mathcal{G}^{\mathrm{germ}}_{\sigma}$, *a compact convex germ body*

$$\mathcal{B}^{\mathrm{germ}}_{\sigma} \subset \mathcal{G}^{\mathrm{germ}}_{\sigma}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{germ}} : \mathcal{G}_\sigma^{\mathrm{germ}} \to \mathcal{S}_\sigma^{\mathrm{seed}},$$

and smooth germ fiber minimizers

$$\mathfrak{f}_\sigma^{\mathrm{germ,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{germ,fdn}}, \quad \mathfrak{f}_\sigma^{\mathrm{germ,grd}}, \quad \mathfrak{f}_\sigma^{\mathrm{germ,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{germ,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{germ,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{germ}}$$

satisfying

$$\Pi_\sigma^{\mathrm{germ}}(\mathcal{B}_\sigma^{\mathrm{germ}}) = \mathcal{B}_\sigma^{\mathrm{seed}}, \qquad \Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,seed}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{seed}}},$$

together with the transport identities

$$\Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,fdn}} = \mathfrak{f}_\sigma^{\mathrm{seed,fdn}}, \qquad \Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,grd}} = \mathfrak{f}_\sigma^{\mathrm{seed,grd}},$$

$$\Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,orig}} = \mathfrak{f}_\sigma^{\mathrm{seed,orig}}, \qquad \Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,prec}} = \mathfrak{f}_\sigma^{\mathrm{seed,prec}},$$

$$\Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,lift}} = \mathfrak{f}_\sigma^{\mathrm{seed,lift}}, \qquad \Pi_\sigma^{\mathrm{germ}} \circ \mathfrak{m}_\sigma^{\mathrm{germ}} = \mathfrak{m}_\sigma^{\mathrm{seed}}.$$

2. *A smooth germ restriction section*

$$\Gamma_\sigma^{\mathrm{germ}} : \mathcal{G}_\sigma^{\mathrm{germ}} \to E_\sigma^{\mathrm{sub}},$$

the germ functional

$$\mathscr{W}_\sigma^{\mathrm{germ}}(g) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\big(\Gamma_\sigma^{\mathrm{germ}}(g), \Gamma_\sigma^{\mathrm{germ}}(g)\big),$$

and the exact germ shell

$$\mathcal{Z}_\sigma^{\mathrm{germ}} \equiv \big(\Gamma_\sigma^{\mathrm{germ}}\big)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{germ}},$$

satisfying

$$\Gamma_\sigma^{\mathrm{germ}} \circ \mathfrak{f}_\sigma^{\mathrm{germ,seed}} = \Gamma_\sigma^{\mathrm{seed}}, \qquad \Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}}.$$

3. *A closed embedded germ zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{germ}} \subset \mathcal{Z}_\sigma^{\mathrm{germ}},$$

the same hidden fiber space $\mathcal{H}_\sigma^{\mathrm{hid}}$ *with the same compact hidden body* $D_\sigma^{\mathrm{hid}} \subset \mathcal{H}_\sigma^{\mathrm{hid}}$*, a diffeomorphism*

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{germ}}} : \mathcal{Z}_\sigma^{0,\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{0,\mathrm{seed}},$$

and a split diffeomorphism

$$\Phi_\sigma^{\mathrm{germ}} : \mathcal{Z}_\sigma^{0,\mathrm{germ}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}$$

obeying the projected split identity

$$\Phi_\sigma^{\mathrm{seed}}(z,\eta) = \Pi_\sigma^{\mathrm{germ}}\Big(\Phi_\sigma^{\mathrm{germ}}\big((\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{germ}}})^{-1}(z), \eta\big)\Big)$$

for every $z \in \mathcal{Z}_\sigma^{0,\mathrm{seed}}$ *and every* $\eta \in D_\sigma^{\mathrm{hid}}$*.*

4. *A hidden charge*

$$\mathcal{Q}_\sigma^{\mathrm{germ}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$

a smooth observer-compatible exact germ zero-response realization

$$\zeta_\sigma^{\mathrm{germ}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{germ}},$$

satisfying

$$\mathcal{Q}_\sigma^{\mathrm{seed}} = \mathcal{Q}_\sigma^{\mathrm{germ}}, \qquad \zeta_\sigma^{\mathrm{seed}} = \Pi_\sigma^{\mathrm{germ}} \circ \zeta_\sigma^{\mathrm{germ}},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}_\sigma^{0,\mathrm{germ}}$, recorded through $\widetilde{K}_\sigma^{0,\mathrm{germ}}$ and $P_\sigma^{\mathrm{germ,co}}$, whose projection through

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{germ}}}$$

recovers the recorded seed kernel chart $\widetilde{K}_\sigma^{0,\mathrm{seed}}$ and orthogonal projector $P_\sigma^{\mathrm{seed,co}}$.

Definition 5 (Recorded germ-generating substrate proto-germ dynamics)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. The recorded proto-germ package $\mathfrak{P}^{\mathrm{proto}}$ for sector σ consists of the following data.*

1. *A finite-dimensional Euclidean proto-germ state space $\mathcal{U}_\sigma^{\mathrm{proto}}$, a compact convex proto-germ body*

$$\mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{U}_\sigma^{\mathrm{proto}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{proto}} : \mathcal{U}_\sigma^{\mathrm{proto}} \to \mathcal{G}_\sigma^{\mathrm{germ}},$$

and smooth proto-germ fiber minimizers

$$\mathfrak{f}_\sigma^{\mathrm{proto,germ}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,fdn}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,grd}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{proto,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{proto}}$$

satisfying

$$\Pi_\sigma^{\mathrm{proto}}(\mathcal{B}_\sigma^{\mathrm{proto}}) = \mathcal{B}_\sigma^{\mathrm{germ}}, \qquad \Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,germ}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{germ}}},$$

together with the transport identities

$$\Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,seed}} = \mathfrak{f}_\sigma^{\mathrm{germ,seed}}, \qquad \Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,fdn}} = \mathfrak{f}_\sigma^{\mathrm{germ,fdn}},$$

$$\Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,grd}} = \mathfrak{f}_\sigma^{\mathrm{germ,grd}}, \qquad \Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,orig}} = \mathfrak{f}_\sigma^{\mathrm{germ,orig}},$$

$$\Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,prec}} = \mathfrak{f}_\sigma^{\mathrm{germ,prec}}, \qquad \Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,lift}} = \mathfrak{f}_\sigma^{\mathrm{germ,lift}},$$

$$\Pi_\sigma^{\mathrm{proto}} \circ \mathfrak{m}_\sigma^{\mathrm{proto}} = \mathfrak{m}_\sigma^{\mathrm{germ}}.$$

2. *A smooth proto-germ restriction section*

$$\Gamma_\sigma^{\mathrm{proto}} : \mathcal{U}_\sigma^{\mathrm{proto}} \to E_\sigma^{\mathrm{sub}},$$

the proto-germ functional

$$\mathscr{W}_\sigma^{\mathrm{proto}}(p) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\big(\Gamma_\sigma^{\mathrm{proto}}(p), \Gamma_\sigma^{\mathrm{proto}}(p)\big),$$

and the exact proto-germ shell

$$\mathcal{Z}_\sigma^{\mathrm{proto}} \equiv (\Gamma_\sigma^{\mathrm{proto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{proto}},$$

satisfying

$$\Gamma_\sigma^{\mathrm{proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,germ}} = \Gamma_\sigma^{\mathrm{germ}}, \qquad \Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

3. *A closed embedded proto-germ zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{proto}} \subset \mathcal{Z}_\sigma^{\mathrm{proto}},$$

the same hidden fiber space $\mathcal{H}_\sigma^{\mathrm{hid}}$ with the same compact hidden body $D_\sigma^{\mathrm{hid}} \subset \mathcal{H}_\sigma^{\mathrm{hid}}$, a diffeomorphism

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{proto}}} : \mathcal{Z}_\sigma^{0,\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{0,\mathrm{germ}},$$

and a split diffeomorphism

$$\Phi_\sigma^{\mathrm{proto}} : \mathcal{Z}_\sigma^{0,\mathrm{proto}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}$$

obeying the projected split identity

$$\Phi_\sigma^{\mathrm{germ}}(z,\eta) = \Pi_\sigma^{\mathrm{proto}}\Big(\Phi_\sigma^{\mathrm{proto}}\big((\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{proto}}})^{-1}(z),\eta\big)\Big)$$

for every $z \in \mathcal{Z}_\sigma^{0,\mathrm{germ}}$ and every $\eta \in D_\sigma^{\mathrm{hid}}$.

4. *A hidden charge*

$$\mathcal{Q}_\sigma^{\mathrm{proto}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$

a smooth observer-compatible exact proto-germ zero-response realization

$$\zeta_\sigma^{\mathrm{proto}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{proto}},$$

satisfying

$$\mathcal{Q}_\sigma^{\mathrm{germ}} = \mathcal{Q}_\sigma^{\mathrm{proto}}, \qquad \zeta_\sigma^{\mathrm{germ}} = \Pi_\sigma^{\mathrm{proto}} \circ \zeta_\sigma^{\mathrm{proto}},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}_\sigma^{0,\mathrm{proto}}$, recorded through $\widetilde{K}_\sigma^{0,\mathrm{proto}}$ and $P_\sigma^{\mathrm{proto,co}}$, whose projection through

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{proto}}}$$

recovers the recorded germ kernel chart $\widetilde{K}_\sigma^{0,\mathrm{germ}}$ and orthogonal projector $P_\sigma^{\mathrm{germ,co}}$.

Remark 2. *Definitions 4 and 5 are the fixed targets already carried by the current proto-germ-frontier theorem. The present note does not spend its move on a second germ-level or proto-germ-level restatement. It records those current packages only because the deeper pre-proto-germ dynamics are defined relative to them.*

4. Proto-Germ-Generating Substrate Pre-Proto-Germ Dynamics

Definition 6 (Proto-germ-generating substrate pre-proto-germ dynamics)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. A proto-germ-generating substrate pre-proto-germ dynamics package $\mathfrak{Q}^{\mathrm{preproto}}$ for sector σ consists*

of the following data.

1. *A finite-dimensional Euclidean pre-proto-germ state space* $\mathcal{V}_\sigma^{\mathrm{preproto}}$*, a compact convex pre-proto-germ body*

$$\mathcal{B}_\sigma^{\mathrm{preproto}} \subset \mathcal{V}_\sigma^{\mathrm{preproto}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to \mathcal{U}_\sigma^{\mathrm{proto}},$$

and smooth pre-proto-germ fiber minimizers

$$\mathfrak{f}_\sigma^{\mathrm{preproto,proto}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,germ}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}},$$
$$\mathfrak{f}_\sigma^{\mathrm{preproto,grd}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{preproto}}$$

satisfying

$$\Pi_\sigma^{\mathrm{preproto}}(\mathcal{B}_\sigma^{\mathrm{preproto}}) = \mathcal{B}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}},$$

together with the transport identities

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,germ}} = \mathfrak{f}_\sigma^{\mathrm{proto,germ}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,seed}} = \mathfrak{f}_\sigma^{\mathrm{proto,seed}},$$
$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}} = \mathfrak{f}_\sigma^{\mathrm{proto,fdn}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,grd}} = \mathfrak{f}_\sigma^{\mathrm{proto,grd}},$$
$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,orig}} = \mathfrak{f}_\sigma^{\mathrm{proto,orig}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,prec}} = \mathfrak{f}_\sigma^{\mathrm{proto,prec}},$$
$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,lift}} = \mathfrak{f}_\sigma^{\mathrm{proto,lift}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{m}_\sigma^{\mathrm{preproto}} = \mathfrak{m}_\sigma^{\mathrm{proto}}.$$

2. *A smooth pre-proto-germ restriction section*

$$\Gamma_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to E_\sigma^{\mathrm{sub}},$$

the pre-proto-germ functional

$$\mathscr{W}_\sigma^{\mathrm{preproto}}(q) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}(\Gamma_\sigma^{\mathrm{preproto}}(q), \Gamma_\sigma^{\mathrm{preproto}}(q)),$$

and the exact pre-proto-germ shell

$$\mathcal{Z}_\sigma^{\mathrm{preproto}} \equiv (\Gamma_\sigma^{\mathrm{preproto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{preproto}},$$

satisfying

$$\Gamma_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \Gamma_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} : \mathcal{Z}_\sigma^{\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}.$$

3. *A closed embedded pre-proto-germ zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{preproto}} \subset \mathcal{Z}_\sigma^{\mathrm{preproto}},$$

the same hidden fiber space $\mathcal{H}_\sigma^{\mathrm{hid}}$ *with the same compact hidden body* $D_\sigma^{\mathrm{hid}} \subset \mathcal{H}_\sigma^{\mathrm{hid}}$*, a diffeo-*

morphism

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{0,\mathrm{proto}},$$

and a split diffeomorphism

$$\Phi_\sigma^{\mathrm{preproto}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}}$$

obeying the projected split identity

$$\Phi_\sigma^{\mathrm{proto}}(z,\eta) = \Pi_\sigma^{\mathrm{preproto}}\Big(\Phi_\sigma^{\mathrm{preproto}}\big((\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}})^{-1}(z),\eta\big)\Big)$$

for every $z \in \mathcal{Z}_\sigma^{0,\mathrm{proto}}$ *and every* $\eta \in D_\sigma^{\mathrm{hid}}$.

4. *A hidden charge*

$$\mathcal{Q}_\sigma^{\mathrm{preproto}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$

a smooth observer-compatible exact pre-proto-germ zero-response realization

$$\zeta_\sigma^{\mathrm{preproto}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{preproto}},$$

satisfying

$$\mathcal{Q}_\sigma^{\mathrm{proto}} = \mathcal{Q}_\sigma^{\mathrm{preproto}}, \qquad \zeta_\sigma^{\mathrm{proto}} = \Pi_\sigma^{\mathrm{preproto}} \circ \zeta_\sigma^{\mathrm{preproto}},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}_\sigma^{0,\mathrm{preproto}}$*, recorded through* $\widetilde{K}_\sigma^{0,\mathrm{preproto}}$ *and* $P_\sigma^{\mathrm{preproto,co}}$*, whose projection through*

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}}$$

recovers the recorded proto-germ kernel chart $\widetilde{K}_\sigma^{0,\mathrm{proto}}$ *and orthogonal projector* $P_\sigma^{\mathrm{proto,co}}$.

The package is benchmark faithful *if it preserves the benchmark outputs* (1)*, introduces no second operative metric, introduces no enlarged low-energy matter law, and is presented as a deeper substrate-side source for the current proto-germ package rather than as a replacement benchmark theory.*

Remark 3. *The label* pre-proto-germ *is used here only as a local marker for a deeper layer below the current proto-germ frontier. It does not by itself assert a completed microscopic ontology or a final primitive law. Its role is only to name the next sharper transport package below the current proto-germ dynamics.*

5. Projected Exact-Shell Transport to the Proto-Germ Package

Proposition 1 (The proto-germ shell is recovered by projected exact-shell transport)**.** *Fix a benchmark-faithful pre-proto-germ package. Then projected exact-shell transport along*

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} \qquad \textit{and} \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}}$$

recovers the exact proto-germ shell, the embedded proto-germ zero core, and the proto-germ split of Definition 5.

Proof. Definition 6(2) states that the restriction of $\Pi_\sigma^{\mathrm{preproto}}$ to $\mathcal{Z}_\sigma^{\mathrm{preproto}}$ is a diffeomorphism onto $\mathcal{Z}_\sigma^{\mathrm{proto}}$. Definition 6(3) states that the restriction of $\Pi_\sigma^{\mathrm{preproto}}$ to $\mathcal{Z}_\sigma^{0,\mathrm{preproto}}$ is a diffeomorphism onto

$\mathcal{Z}_\sigma^{0,\mathrm{proto}}$ and that the projected image of the pre-proto-germ split is exactly the recorded proto-germ split. These are precisely the claimed projected exact-shell transport statements. □

Proposition 2 (Pre-proto-germ dynamics canonically reproduce the current proto-germ package)**.** *Fix a benchmark-faithful pre-proto-germ package. Then the projected exact-shell transport of Definition 6 reproduces exactly the current germ-generating substrate proto-germ dynamics package of Definition 5.*

Proof. The proto-germ body and the lower minimizer hierarchy are recovered by Definition 6(1): the pre-proto-germ body projects onto the recorded proto-germ body, the pre-proto-germ proto minimizer sections the projection on that body, and the projected pre-proto-germ lower minimizers are exactly the recorded proto-germ minimizers on the fixed germ, seed, foundation, ground, origin, precursor, lift, and substrate fibers.

The proto-germ restriction data are recovered by Definition 6(2): the recorded proto-germ section is the minimizing transport of the pre-proto-germ section, the proto-germ functional is therefore the projected restriction functional, and Proposition 1 recovers the exact proto-germ shell itself.

The remaining proto-germ shell-internal data are recovered by Definition 6(3)–(4): the embedded proto-germ zero core is the projected pre-proto-germ zero core, the proto-germ split is the projected pre-proto-germ split, the proto-germ hidden charge is the pre-proto-germ hidden charge, the exact proto-germ zero-response realization is the projected pre-proto-germ realization, and the fixed-data solvability / coercive package is the projection of the corresponding pre-proto-germ package through the zero-core diffeomorphism. These are exactly the constituents retained by the current proto-germ frontier. □

Proposition 3 (The benchmark package remains fixed)**.** *For every benchmark-faithful pre-proto-germ package, the operative metric remains exactly $g^\sharp$, the ordinary matter route remains unchanged, no second operative metric is introduced, and the low-energy matter law is not enlarged.*

Proof. This is part of the benchmark-faithfulness requirement in Definition 6. Equation (1) remains unchanged on the full primitive forcing shell, so the deeper pre-proto-germ package does not alter the benchmark exact-closure package. □

6. Main Theorem

Theorem 1 (Germ-generating substrate proto-germ dynamics from proto-germ-generating substrate pre-proto-germ dynamics)**.** *Fix the primitive continuation data of Definition 1, the recorded lower-fiber ladder of Definition 2, the recorded downstream seed package of Definition 3, the recorded germ package of Definition 4, and the recorded proto-germ package of Definition 5. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, every benchmark-faithful proto-germ-generating substrate pre-proto-germ dynamics package canonically produces the current proto-germ package as projected exact-shell transport from the deeper pre-proto-germ layer. More precisely:*

1. *the current germ-generating substrate proto-germ dynamics are reproduced unchanged, sector by sector, by projected exact-shell transport from the deeper pre-proto-germ package;*

2. *because the induced proto-germ package is exactly the one already used by the current proto-germ-frontier theorem, the current germ package, the current seed package, the minimal root*

exact-shell transport nucleus, and the previously recorded downstream root-side consequences remain available unchanged;

3. *the same benchmark one-metric package remains fixed: the operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged;*

4. *consequently the remaining honest burden on the active line is deeper again: derive or justify the proto-germ-generating substrate pre-proto-germ dynamics themselves from still more primitive substrate structure, or else prove a genuinely smaller theorem-level collapse at that pre-proto-germ level.*

Proof. Item (1) is Proposition 2. Item (2) is the routing consequence of item (1): the present note reproduces exactly the proto-germ package that the current proto-germ-frontier theorem already uses as its input, so that theorem and its downstream recovery of the germ package, the seed package, and the minimal root exact-shell transport nucleus apply unchanged. Item (3) is Proposition 3. Item (4) is the project-level consequence of the previous items. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 1, the remaining honest burden is no longer to justify the germ-generating substrate proto-germ dynamics themselves. Those dynamics are now the canonical projected exact-shell output of a sharper pre-proto-germ class. What remains open is why the proto-germ-generating substrate pre-proto-germ dynamics should hold, or whether an even sharper theorem can remove that still deeper pre-proto-germ-level freedom while preserving the same benchmark one-metric package and claim boundary.*

Proof. Theorem 1 removes the current proto-germ package as the unexplained stopping point on the active line. The next honest burden therefore lies strictly below it, unless a smaller theorem-level core inside the pre-proto-germ package can first be isolated and proved sufficient. □

7. Conclusion

The positive result of this note is narrower than a completed first-principles derivation and stronger than another same-output proto-germ restatement. The current proto-germ-frontier theorem had already shown that the present germ package is the exact-shell transport image of a more primitive proto-germ package. The present theorem then removes the next unexplained stopping point by showing that this proto-germ package itself is the projected exact-shell transport image of a more primitive proto-germ-generating substrate pre-proto-germ dynamics.

The scope remains disciplined. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to completed derivation or licenses stronger promotion language.

The open burden therefore moves one honest step further again. The next benchmark-facing manuscript should derive or justify the pre-proto-germ dynamics from still more primitive substrate structure, unless the sharper honest gain available instead is a genuinely smaller theorem-level collapse inside the observer-relevant pre-proto-germ package. Until then, the present result should be read for what it is: a deeper proto-germ-scope derivation on the active line of Æther-flow Interpretation of Relativity, not a claim that the full first-principles program is finished.